Cluster-Adaptive Prompting for Multi-Turn Travel and Tourism Conversational Systems

Final Report

Surnella Sandeep

Department of Data Analytics, Walsh College

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Mr. Smitraj Raut

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GitHub repository: https://github.com/surnella/QM640-Dialogue-Clustering-Capstone

Contents

Abstract 4

Introduction 5

Background and Context 5

Problem Statement 6

Purpose of the Study and Research Questions 7

Contributions and Expected Value 8

Literature Review 8

Literature Review Approach and Relevance Matrix 8

Thematic Synthesis and Research Gaps 9

Materials and Methods 11

Data Sources, Access, and Ethical Governance 11

Data Dictionary and Data Cleaning 11

Exploratory Data Analysis (Univariate, Bivariate, and Multivariate) 13

Research Design, Hypotheses, and Statistical Power Analysis 16

Model Selection, Mathematical Formulation, and Evaluation Metrics 17

Architecture and Workflow 19

System Architecture and Algorithmic Workflows 19

Pipeline Component Specifications and Technology Stack 20

Results, Analysis, and Synthesis 21

Consolidated Model Performance and Hypothesis Testing Summary 21

RQ1 Results: Trajectory Clustering, Refined Archetypes, and Prefix Routing 22

RQ2 Results: Cluster-Adaptive Few-Shot Prompting on Primary Versus Refined Archetypes 25

RQ3 Results: 2 × 2 Factorial Evaluation of Declarative Rules and Adaptive Prompting .27

RQ4 Results: 2 × 3 Multi-Turn Simulation of Turn Economy Across Archetypes 29

Technical Interpretation, Business Significance, and Synthesis of Conclusions.....31

Implementation, Limitations, and Future Scope.....32

Deployment Architecture, System Integration, and Real-World Use Case.....32

Computational Reproducibility and Operational Requirements.....33

Study Limitations and Threats to Validity.....33

Future Scope and Research Extensions 33

References.....35

Appendix A.....40

Appendix B 43

Appendix C 44

Appendix D.....47

Appendix E 51

Abstract

Online travel agencies increasingly deploy large language model (LLM) assistants for package booking, yet static prompts ignore traveler decision-making heterogeneity, while naive few-shot prompting leaks stale entities and violates booking policies. This study evaluates a neuro-symbolic, cluster-adaptive framework coupling early-turn behavioral-semantic archetype routing and slot-masked FAISS retrieval with a declarative rule engine. Using 1,369 Microsoft Frames dialogues (1,334 analytic: golden $n = 933$, test $n = 401$) and 1,499 MultiWOZ 2.2 dialogues, four hypotheses were tested with Sentence-BERT, Gemini 3,072-d embeddings, Gemini 2.5 Flash, and Gemini 2.5 Pro. For RQ1, clustering algorithms differed across 159 runs, $F(2, 156) = 91.33$, $p < .001$, $\eta^2 = .54$, and combining Gemini prefix embeddings with a 16-field behavioral profile on the first nine user messages yielded three stable refined archetypes—*Decisive Booker*, *Open Explorer*, and *Rigid-Constraint Seeker* (ARI = .76; held-out routing $\kappa = .535$ at $M = 5$ to .958 at $M = 9$). For RQ2 ($N = 33$ repeated-measures), cluster-adaptive prompting increased style convergence, $F(1.88, 60.29) = 9.79$, $p < .001$, $\eta_p^2 = .23$, and cut pacing error to 3.87 words, $F = 7.10$, $p = .007$. For RQ3 ($N = 144$ 2 × 2 factorial; dual-judge $\kappa = .917$), active rules raised violation enforcement from 5.6% to 98.6%, $F(1, 140) = 43.92$, $p < .001$, eliminated 35.9% entity leakage to 0.0%, interaction $F(1, 140) = 34.82$, $p < .001$, and raised slot F_1 to .952, $F(1, 140) = 489.52$, $p < .001$. For RQ4 ($N = 180$ 2 × 3 simulation; persona $\kappa = .933$), cluster-adaptive prompting cut turns by 29.5%–42.7%, $F(1, 174) = 664.43$, $p < .001$, interaction $F(2, 174) = 10.89$, $p < .001$, and eliminated redundant slot elicitation, $F(1, 174) = 63.02$, $p < .001$.

Keywords: conversational AI, travel chatbots, trajectory clustering, few-shot learning, guardrails

Introduction

Background and Context

Conversational artificial intelligence (AI) assistants have become a core digital channel in travel, tourism, and hospitality, supporting destination discovery, package comparison, and itinerary booking (Buhalis & Sinarta, 2019; Ivanov & Webster, 2019; Jannach et al., 2021). Unlike single-turn question answering, vacation package booking is a consultative, multi-turn negotiation in which travelers balance competing constraints across origin, destination, dates, duration, hotel tier, and budget (El Asri et al., 2017). Empirical analyses of human travel dialogues demonstrate that travelers converse in distinct behavioral styles: some state fixed parameters up front and seek immediate confirmation, others explore multiple destinations and hotel tiers side by side, and others insist on rigid date and luxury constraints that require proactive trade-off resolution when preferred inventory exceeds their budget ceiling (Inaba et al., 2022; Melián-González et al., 2021).

Despite this behavioral diversity, commercial large language model (LLM) travel assistants typically use one static system prompt and a uniform sequential intake script. Asking clarifying questions one slot at a time or offering a single hotel option per turn creates repetitive turns and session abandonment. Conversely, adapting an LLM's style purely through in-context few-shot examples introduces two severe reliability hazards (Liu et al., 2022; Rebedea et al., 2023): first, unmasked retrieved examples cause factual entity leakage, where the model copies stale hotel names, airfares, or promo codes from the examples into the active session; second, stylistic prompt instructions compete with business rules, causing the assistant to overlook mandatory visa windows, passport validity thresholds, party-size limits, or discount caps (Yao et

al., 2024). Resolving both personalization and compliance requires a neuro-symbolic architecture that decouples behavioral style adaptation from deterministic policy verification.

Problem Statement

Business problem statement. Online travel agencies (OTAs), tour operators, and hospitality contact centers suffer from low self-service conversion and elevated session abandonment because static conversational assistants fail to adapt their information density and recommendation strategy to distinct customer decision-making styles, while unverified generative responses expose enterprises to financial and regulatory liability from booking policy violations and hallucinated package terms (Buhalis & Sinarta, 2019; Ivanov & Webster, 2019; Melián-González et al., 2021).

Technical problem statement. Existing LLM dialogue systems lack (a) an empirically validated representation for clustering multi-turn user behavior into stable archetypes that can be routed accurately from early dialogue prefixes without end-of-dialogue outcome leakage, (b) a slot-masked exemplar retrieval mechanism that achieves target archetype style convergence without inducing entity leakage, and (c) formal factorial evidence quantifying the main and interaction effects of external declarative rule verification and cluster-adaptive prompting on slot fidelity, policy adherence, and multi-turn turn economy (Acikgoz et al., 2025; Liu et al., 2022; Rebedea et al., 2023).

Measurable research formulation. This study models six quantitative dependent variables (Y)—cluster separation Silhouette (Y_1), style convergence cosine and pacing error (Y_2), policy rule adherence rate (Y_3), exemplar entity leakage rate (Y_4), factual slot F_1 (Y_5), and log-transformed turns to goal completion and redundant elicitation rate (Y_6)—as functions of four manipulated independent variables (X): clustering algorithm and trajectory representation (X_1,

RQ1), in-context conditioning strategy (X_2, RQ2), rule-engine mode crossed with prompting strategy (X_3, RQ3), and prompting strategy crossed with refined user archetype (X_4, RQ4).

Purpose of the Study and Research Questions

The purpose of this quantitative study is to develop and evaluate an end-to-end neuro-symbolic, cluster-adaptive conversational framework on public human travel dialogues across four linked research questions (Table 1).

Table 1

Research Questions, Hypotheses, and Analytical Framework

RQ	Research question	Null and alternative hypotheses	Experimental design and statistical test
RQ1	Which clustering algorithm (K-Means, Ward agglomerative, HDBSCAN) best separates dialogue embeddings, and which stable, early-routable user archetypes emerge?	H_{01}: Equal mean Silhouette across algorithms. H_{a1}: At least one differs, and truncated behavioral-semantic prefixes yield stable (ARI ≥ .60), routable archetypes.	One-way ANOVA across N = 159 runs (Welch & Kruskal–Wallis checks; Tukey & Games–Howell post hoc) + representation × truncation grid
RQ2	Does cluster-adaptive few-shot prompting increase style convergence and pacing calibration relative to zero-shot and static few-shot prompting?	H_{02}: Equal mean style cosine across conditions. H_{a2}: Cluster-adaptive > static and zero-shot on refined user archetypes.	One-way repeated-measures ANOVA (N = 33 dialogues × 3 conditions = 99 calls; Greenhouse–Geisser, Friedman, Holm contrasts)
RQ3	How do a declarative rule engine and cluster-adaptive prompting interact to govern rule adherence, exemplar leakage, slot F_1, and style alignment?	H_{03}: No main or interaction effects. H_{a3}: Active rules raise adherence and slot F_1 and eliminate few-shot entity leakage.	2 × 2 factorial ANOVA (N = 144, n = 36/cell; OLS, HC3, and ART checks; dual-LLM judge κ)
RQ4	Does cluster-adaptive prompting with an informed-slot ledger reduce turns to goal completion and redundant elicitation differently across user archetypes?	H_{04}: No prompting or prompting × archetype interaction effects. H_{a4}: Cluster-adaptive prompting reduces log turns and RER with an archetype interaction.	2 × 3 factorial ANOVA on log turns and RER (N = 180, n = 30/cell; OLS, HC3, ART, Welch t, Mann–Whitney U; persona κ)

Note. RQ = research question; ARI = adjusted Rand index; HC3 = heteroscedasticity-consistent SE; ART = aligned rank transform; RER = redundant elicitation rate. All tests use α = .05.

Contributions and Expected Value

This study delivers four contributions: (1) proving that truncating dialogues to the first nine user messages and combining Gemini prefix embeddings with structured LLM behavioral ratings yields stable user archetypes (ARI = .76) with high held-out prefix routing agreement (κ = .535 at five messages to .958 at nine messages); (2) contrasting primary versus refined archetypes in RQ2 to demonstrate that cluster-adaptive prompting succeeds significantly ($p <$ .001) once upstream routing is coherent; (3) demonstrating in RQ3 that external declarative verification eliminates the 35.9% entity leakage of unguarded few-shot prompting while raising policy violation enforcement from 5.6% to 98.6%; and (4) showing in RQ4 that archetype-adaptive turn-economy policies cut turns to booking completion by 29.5% to 42.7% and eliminate redundant slot questions.

Literature Review

Literature Review Approach and Relevance Matrix

Peer-reviewed studies were retrieved from the ACL Anthology, ACM Digital Library, IEEE Xplore, ScienceDirect, Google Scholar, and arXiv using Boolean combinations of *travel and hospitality chatbots*, *conversational recommender systems*, *task-oriented dialogue corpora*, *trajectory clustering*, *in-context exemplar retrieval*, and *LLM guardrails*. Twenty-six sources spanning business adoption, dialogue datasets, unsupervised clustering, retrieval-augmented prompting, and statistical methodology were synthesized. Table 2 maps the core literature streams to the gaps addressed in RQ1–RQ4.

Table 2

Literature Relevance Matrix Across Business and Technical Domains

Study and stream	Methodology and dataset	Key findings	Gap addressed in this study
Buhalis & Sinarta (2019); Ivanov & Webster (2019); Melián-González et al. (2021) — Travel AI business value	Service-automation frameworks and consumer adoption modeling (N = 1,476) in tourism and hospitality	Real-time personalization and low interaction effort drive chatbot booking conversion and ROI	Measures perceptions via surveys rather than instrumenting turn economy, slot redundancy, or policy compliance in LLM dialogues (RQ3, RQ4)
Jannach et al. (2021); Inaba et al. (2022) — Conversational recommenders & traveler styles	CRS architectural taxonomy; collection and analysis of multi-turn travel agency dialogues	Travelers differ in initiative, exploration breadth, and constraint rigidity; static policies cause repetitive turns	Describes traveler diversity qualitatively without unsupervised archetype clustering or prefix routing (RQ1, RQ2)
El Asri et al. (2017); Budzianowski et al. (2018); Zang et al. (2020) — Dialogue corpora	Microsoft Frames (1,369 package negotiations) and MultiWOZ 2.2 (10,438 multi-domain dialogues)	Frames captures nonlinear frame-switching and budget trade-offs; MultiWOZ standardizes slot tracking	Built for supervised state tracking; neither clusters user archetypes nor evaluates few-shot leakage or policy rules (RQ1–RQ4)
Reimers & Gurevych (2019); Lee et al. (2025); McInnes et al. (2017, 2018); Hennig (2007) — Embeddings & clustering	Sentence-BERT (384-d), Gemini embeddings (3,072-d), HDBSCAN, UMAP, and bootstrap ARI stability	Cosine geometry supports semantic clustering; bootstrap ARI and Silhouette quantify partition stability	Clusters static texts rather than prefix-truncated user trajectories paired with LLM behavioral ratings (RQ1)
Brown et al. (2020); Liu et al. (2022); Johnson et al. (2021) — In-context learning & FAISS	Few-shot prompting; k-NN exemplar selection (KATE); FAISS vector indexing	Nearest-neighbor in-context examples outperform random static examples on single-turn NLP tasks	Retrieves from one global pool without archetype partitioning, slot masking, or style/leakage evaluation (RQ2, RQ3)
Rebedea et al. (2023); Yao et al. (2024); Acikgoz et al. (2025); Gilardi et al. (2023) — Guardrails & evaluation	NeMo Guardrails; τ-bench policy benchmark; TD-EVAL metrics; LLM annotation reliability	Prompt-only LLMs violate domain rules; programmable rails and turn-level factual audits are essential	Does not test the 2 × 2 factorial interaction between declarative rules and cluster-adaptive style prompting (RQ3)

Note. CRS = conversational recommender system; ARI = adjusted Rand index; FAISS = Facebook AI Similarity Search.

Thematic Synthesis and Research Gaps

Business context: Hospitality service automation and conversational friction. In digital travel and hospitality commerce, competitive differentiation depends on real-time itinerary co-creation ('nowness' service; Buhalis & Sinarta, 2019). Ivanov and Webster (2019) showed that

automated service agents generate financial value only when they resolve consultative inquiries without human escalation, while Melián-González et al. (2021) demonstrated that consumer intention to use travel chatbots depends on interaction efficiency and task fit. Empirical studies of travel agency dialogues confirm that human consultants adapt their pacing—confirming bookings rapidly for decisive buyers while presenting side-by-side comparisons for exploratory shoppers (Inaba et al., 2022; Jannach et al., 2021).

Task-oriented corpora and unsupervised trajectory clustering. While MultiWOZ (Budzianowski et al., 2018) and MultiWOZ 2.2 (Zang et al., 2020) benchmark slot tracking across local services, they lack flight routing and multi-package price negotiation. Microsoft Frames (El Asri et al., 2017) directly captures multi-turn flight-and-hotel negotiation under budget constraints, making it the ideal primary corpus and MultiWOZ 2.2 an external structural comparison. Representing dialogues with Sentence-BERT (Reimers & Gurevych, 2019) and Gemini embeddings (Lee et al., 2025) enables rigorous comparison of K-Means, Ward agglomerative clustering, and HDBSCAN (McInnes et al., 2017) using Silhouette (Rousseeuw, 1987), Davies–Bouldin (Davies & Bouldin, 1979), bootstrap ARI stability (Hennig, 2007; Hubert & Arabie, 1985), and UMAP (McInnes et al., 2018; Pedregosa et al., 2011), augmented by reliable zero-temperature LLM behavioral ratings (Gilardi et al., 2023).

Dynamic in-context learning, neuro-symbolic guardrails, and research gaps. Few-shot prompting (Brown et al., 2020) and similarity-based FAISS retrieval (Johnson et al., 2021; Liu et al., 2022) improve single-turn task accuracy but risk entity contamination and policy drift in multi-turn dialogues. Programmable guardrails (Rebedea et al., 2023), policy benchmarks (τ-bench; Yao et al., 2024), and turn-level evaluation frameworks (TD-EVAL and BERTScore; Acikgoz et al., 2025; Zhang et al., 2020) motivate decoupling style from policy verification. Four

research gaps follow: (1) discovering early-routable travel user archetypes free of outcome confounding (RQ1); (2) testing archetype-matched, slot-masked exemplar retrieval for style convergence (RQ2); (3) isolating the 2 × 2 factorial interaction between declarative rule verification and adaptive prompting (RQ3); and (4) quantifying multi-turn turn economy and redundant elicitation reductions across traveler personas (RQ4).

Materials and Methods

Data Sources, Access, and Ethical Governance

This study uses two public English dialogue corpora (Table 3): Microsoft Frames (El Asri et al., 2017; 1,369 raw dialogues, 1,334 analytic) and MultiWOZ 2.2 (Zang et al., 2020; seeded sample of 1,500 travel dialogues, 1,499 analytic). Both datasets were collected with research consent, use anonymized role-player IDs, and contain no personally identifiable information.

Table 3

Dataset Overview

Attribute	Microsoft Frames (primary corpus)	MultiWOZ 2.2 (comparison corpus)
Records	1,369 raw; 1,334 analytic (golden D_{gold} = 933, held-out D_{test} = 401)	10,438 raw; travel sample of 1,500; 1,499 analytic
Domain setting	Multi-turn flight + hotel package negotiation under budget and date constraints	Multi-domain city services (hotel, train, taxi, attraction) without flights
Annotations used	Utterance text, speaker, dialogue acts, slot–value arguments, wizard task success, user rating	Utterance text, speaker, active services, turn-level slot state dictionaries
Analytical role	RQ1 clustering & refined archetypes, RQ2 prompting, RQ3 2×2 experiment, RQ4 2×3 simulation	External descriptive comparison in EDA and cross-corpus projection check in RQ1

Data Dictionary and Data Cleaning

Both corpora were normalized into a two-tier Pydantic schema (*src/canonical_schema.py*): Tier 1 stores nine universal travel slots (origin, destination,

departure/return dates, duration in nights, adults, children, budget in USD, and hotel tier), and Tier 2 stores five policy slots evaluated by the rule engine (visa window in days, passport validity in months, package type, room sharing, and discount percentage). Table 4 defines the analytical variables and units (raw source fields and canonical schema fields are listed in Appendix A, Table A3). Table 5 documents the programmatic quality audit and cleaning log. To prevent data leakage, the 1,334 analytic Frames dialogues were split once (seed 42, stratified on task success) into a 70% golden discovery partition (D_{gold}, $n = 933$) used to fit clusters, PCA transformations, routers, and FAISS indices, and a 30% held-out test partition (D_{test}, $n = 401$) used strictly for out-of-sample evaluation.

Table 4

Data Dictionary of Analytical Variables With Units and Operational Definitions

Variable	Role	Type and unit	Operational definition and scale	RQ
Clustering algorithm	Primary IV	Categorical (3 levels)	K-Means (k=3), Ward agglomerative (k=3), HDBSCAN	1
Silhouette & Davies–Bouldin	Primary/Sec. DV	Continuous (unitless / ratio)	Cosine Silhouette [−1, 1] and cluster scatter ratio (≥ 0)	1
Adjusted Rand index (ARI)	Diagnostic DV	Continuous [−1, 1] (unitless)	Chance-corrected partition stability across 80% subsamples	1
LLM behavioral profile (16 fields)	Clustering / Audit	10 ordinal (1–5) + 6 cat./bool.	10 Likert scales + 3 goal/tone clustering fields + 3 audit fields (Table C1)	1, 4
Refined user archetype	IV / Moderator	Categorical (3 levels)	*Decisive Booker*, *Open Explorer*, *Rigid-Constraint Seeker*	1–4
Prompting strategy	Primary IV	Categorical (2 or 3 levels)	Zero-Shot, Static Few-Shot (k=3), Cluster-Adaptive Few-Shot (k=3)	2–4
Style convergence & Pacing err.	Primary/Sec. DV	Cosine [−1, 1] / Words (count)	Cosine to target archetype agent centroid; $\lvert \text{words}_{\text{gen}} - \mu_{\text{words},a} \rvert$	2, 3
Rule-engine mode	Primary IV	Binary (2 levels)	Rules Off (unguarded) vs. Rules Active (pre-check + verification)	3
Rule adherence rate (RAR)	Primary DV	Proportion [0, 1] (rate)	Share of violated policy rules enforced (and compliant states unblocked)	3
Exemplar leakage rate (ELR)	Primary DV	Proportion [0, 1] (rate)	Share of response entities originating from prompt exemplars	3
Factual slot coverage F_1	Primary DV	Proportion [0, 1] (F_1 score)	Harmonic mean of precision and recall over active booking state slots	3

Variable	Role	Type and unit	Operational definition and scale	RQ
Turns to goal (log) & RER	Primary/Sec. DV	Log count (ln turns) / Rate [0, 1]	Log turns until [ACCEPT_BOOKING]; redundant slot requests / agent turns	4

Note. IV = independent variable; DV = dependent variable; RER = redundant elicitation rate. Raw and canonical fields are in Table A3.

Table 5

Data Cleaning and Quality Audit Log

Issue	Variables	Detection	Frames	MultiWOZ	Treatment and rationale
Duplicate full text	Utterances	SHA-256 text hash	0	1	Removed duplicate to prevent double-weighting
Fewer than 3 turns	total_turns	Turn count < 3	35	0	Excluded; minimum 3 turns needed for prefix routing
Turn-count outliers	total_turns	> Q3 + 1.5 × IQR	22 (> 32.5)	13 (> 27.0)	Retained as genuine negotiations; log-transformed in RQ4
Unobserved slots	Tier 1 & 2 slots	Never mentioned	Table E1	Table E1	Preserved as null (None); never imputed
Entity values in text	Utterance text	Slot/number regex	All	All	Masked with [SLOT] tokens in exemplars to prevent leakage
Analytic sample	All fields	Post-cleaning count	1,334 (933/401)	1,499	Locked with stratified 70/30 split (seed 42)

Note. Zero duplicate IDs or empty utterances occurred in either corpus. Missing user ratings (11 in Frames) were retained.

Exploratory Data Analysis (Univariate, Bivariate, and Multivariate)

Univariate Analysis

Table 6 and Figure 1 summarize the univariate distributions. Frames and MultiWOZ 2.2 have similar mean lengths (M = 14.93 vs. 15.17 turns), but Frames exhibits greater dispersion (SD = 6.74 vs. 4.46) and a heavier right tail due to multi-offer package comparisons. Golden and held-out partitions are nearly identical across all metrics. Slot completeness (Appendix E, Table E1 and Figure E1) confirms high coverage of origin (98.4%), destination (99.1%), dates (96.0%),

and budget (78.5%) in Frames, but 0.0% coverage of visa, passport, and discount rules in both corpora, motivating synthetic Tier-2 injection in RQ3.

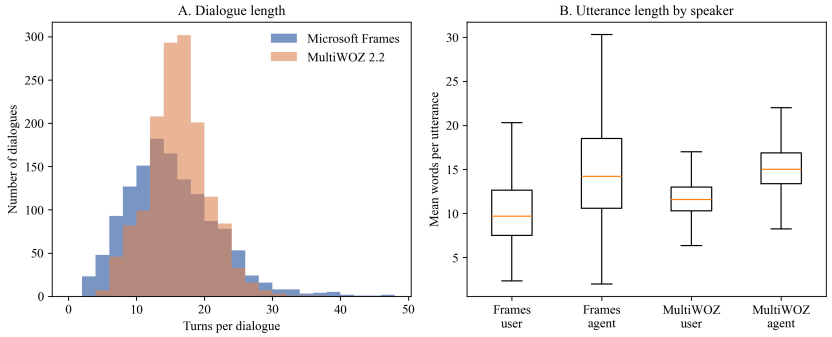
Table 6

Univariate Descriptive Statistics of the Analytic Samples

Sample	*N*	Turns *M* (*SD*)	*Mdn*	Range	User words	Agent words	Task success (%)
Microsoft Frames	1,334	14.93 (6.74)	14	3–48	10.6	15.0	94.2
MultiWOZ 2.2	1,499	15.17 (4.46)	16	4–36	11.7	15.1	—
Frames golden partition (D_gold)	933	14.95 (6.75)	14	3–48	10.7	15.3	94.2
Frames held-out partition (D_test)	401	14.90 (6.71)	14	3–43	10.4	14.5	94.3

Figure 1

Univariate Distribution of Dialogue Length by Corpus



Note. Frames $N = 1{,}334$; MultiWOZ $N = 1{,}499$.

Bivariate Analysis

Parametric (Pearson *r*) and non-parametric (Spearman ρ) bivariate correlations across all 1,334 Frames dialogues (Table 7; Figure 2) show that dialogue length is strongly driven by slot revisions ($r = .74$, $\rho = .76$, $p < .001$) and agent package offers ($r = .56$, $\rho = .61$, $p < .001$). Linking structural counts to LLM behavioral ratings confirms construct validity: slot revisions correlate

strongly with user exploration breadth ($r = .60$, $\rho = .67$, $p < .001$), whereas decisiveness correlates negatively with dialogue length ($r = -.31$, $\rho = -.29$, $p < .001$).

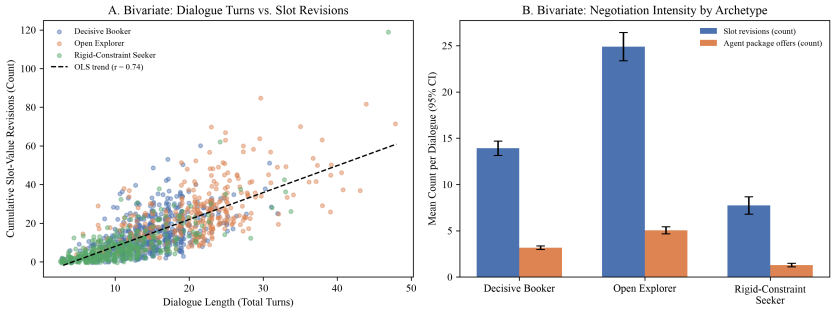
Table 7

Bivariate Parametric and Non-Parametric Correlations of Dialogue Dynamics (Frames)

Variable pair	Pearson *r*	*p* (Pearson)	Spearman ρ	*p* (Spearman)
Total Turns vs. Slot Revisions	.741	< .001	.759	< .001
Total Turns vs. Agent Package Offers	.561	< .001	.608	< .001
Total Turns vs. User Hesitation Markers	.336	< .001	.355	< .001
Slot Revisions vs. Exploration Breadth Rating	.603	< .001	.673	< .001
Total Turns vs. User Decisiveness Rating	−.310	< .001	−.289	< .001
User Words/Turn vs. Information Up Front Rating	.158	< .001	.206	< .001
Agent Offers vs. Destination Flexibility Rating	.324	< .001	.358	< .001

Figure 2

Bivariate Relationships Between Dialogue Length, Slot Revisions, and Archetype Negotiation Intensity



Note. Panel A: Turns vs. slot revisions with OLS trend. Panel B: Mean revisions and package offers by refined archetype (95% CI).

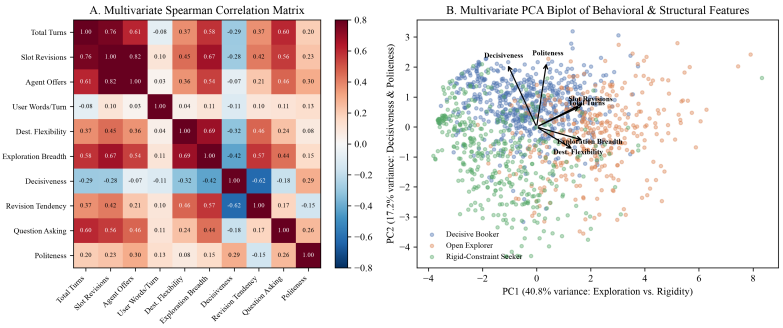
Multivariate Analysis

Multivariate structure across ten structural and behavioral variables was examined via Spearman rank correlation and principal component analysis (PCA; Figure 3; full loadings in Appendix E, Table E2). The first three components explain 68.6% of total variance: PC1

(40.8%%) captures *Exploration vs. Rigidity* (revisions .42, turns .39, exploration breadth .39), PC2 (17.2%%) captures *Decisiveness & Politeness* (−.51, −.46), and PC3 (10.6%%) isolates *User Verbosity* (.89), separating the three refined archetypes across a continuous behavioral manifold.

Figure 3

Multivariate Spearman Correlation Matrix and Principal Component Biplot of Behavioral Features



Note. Panel A: Rank correlation matrix. Panel B: PCA biplot colored by refined archetype with top loading vectors.

Research Design, Hypotheses, and Statistical Power Analysis

Sample sizes were computed a priori for each planned analysis using Cohen's medium effect $f = .25$ (Cohen, 1988), $\alpha = .05$, and target power $1 - \beta \geq .80$ via the noncentral F distribution (Faul et al., 2007):

$$1 - \beta = 1 - F_{nc}(F_{crit}; df_1, df_2, \lambda), \quad F_{crit} = F_{1-\alpha}(df_1, df_2) \quad (1)$$

where $\lambda = f^2 N$ for between-subjects designs (RQ1, RQ3, RQ4) and $\lambda = f^2 N m \varepsilon / (1 - \rho)$ for repeated-measures RQ2 ($m = 3$, $\rho = .50$, $\varepsilon = 1$). Minimum sample sizes were rounded to balanced cells with a 10% buffer (Table 8), yielding final power of .835–.865.

Table 8

Sample Size and Power Calculations Matched to Each Planned Analysis

RQ	Planned analysis	*df* at N_{min}	λ	F_{crit}	N_{min}	Power	N_{final} (per group)	Final power
RQ1	One-way ANOVA (3 algorithms)	2, 155	9.88	3.05	158	.802	159 (53)	.805
RQ2	One-way repeated-measures ANOVA (3 conditions)	2, 54	10.50	3.17	28	.812	33 (11)	.878
RQ3	2 × 2 factorial ANOVA	1, 124	8.00	3.92	128	.801	144 (36)	.846
RQ4	2 × 3 factorial ANOVA	2, 152	9.88	3.06	158	.802	180 (30)	.855

Model Selection, Mathematical Formulation, and Evaluation Metrics

Trajectory and hybrid user representations (RQ1). Primary trajectory vectors pool de-lexicalized turn embeddings $\mathbf{e}_t$ with exponential decay ($\lambda_d = 0.15$):

$$\mathbf{v} = \Sigma_{t=1}^{T} w_t \mathbf{e}_t, \quad w_t = \exp[-\lambda_d(T - t)] \,/\, \Sigma_{j=1}^{T} \exp[-\lambda_d(T - j)] \quad (2)$$

To prevent end-of-dialogue outcome confounding, the refined archetype representation (*Blocks*) concatenates two L2-normalized blocks computed strictly on the first *M* user messages: (1) the top 20 principal components (fitted on D_{gold}) of the 3,072-dimensional Gemini embedding (*gemini-embedding-001*; Lee et al., 2025), and (2) a standardized 16-field LLM behavioral schema comprising 13 clustering fields (ten 1–5 ordinal rating scales plus primary goal, secondary goals, and emotional tone) and three audit-only fields (session outcome, party type, and unusual-conversation flag, excluded from clustering input; Table C1; test–retest weighted κ = .87–.96 across the 10 ordinal scales and κ = .75–.98 across the 6 categorical/boolean fields; revision correlation ρ = .73, Table C2). In RQ2–RQ4, *RefinedArchetypeRouter* projects each test dialogue's opening user prefix into this golden-fitted *Blocks* space, predicts the user archetype via L1-regularized logistic regression, and retrieves *k* = 3 slot-masked exemplars from that archetype's FAISS sub-index. In RQ3, the deterministic rule engine (*src/rule_engine.py*, 14/14

boundary tests passed) evaluates five YAML policy rules (Table 9). Table 10 defines all evaluation metrics.

Table 9

Declarative Policy Rules Used in RQ3

Rule ID and name	Trigger condition	Action	Severity
RULE_VISA_WINDOW: Visa Processing Lead Time	departure_lead_days < 15; is_international = true	block and flag	fatal
RULE_PASSPORT_VALIDITY: Passport Validity Threshold	passport_expiry_months < 6; is_international = true	block and flag	fatal
RULE_PAX_CAPACITY: Maximum Adult Capacity per Single Booking	traveler_count_adults > 8; package_type = 'FIT'	convert to group	warning
RULE_DISCOUNT_THRESHOLD: Maximum Discretionary Sales Discount	discount_applied_percentage > 15.0	clamp discount	fatal
RULE_BUDGET_MINIMUM: Minimum Feasible Per-Pax International Budget	budget stated; budget_amount < 500; is_international = true	prompt realistic budget	warning

Table 10

Evaluation Metrics, Mathematical Formulae, and Interpretation

Metric	Formula	RQ	Interpretation
Silhouette & DB index	$s(i) = [b(i)-a(i)]/\max\{a,b\}$; $DB = (1/k)\Sigma \max(\sigma_i+\sigma_j)/d(\mathbf{c}_i,\mathbf{c}_j)$	1	Higher Silhouette [−1, 1] and lower DB (≥ 0) indicate better separation
ARI & Cohen's κ	$ARI = (RI-\mathrm{E}[RI])/(\max RI-\mathrm{E}[RI])$; $\kappa = (p_o-p_e)/(1-p_e)$	1, 3, 4	Resampling stability (ARI ≥ .60) and routing/judge agreement
Style cosine & Pacing err.	$\cos(\mathbf{r}, \mathbf{c}_a) = (\mathbf{r}\cdot\mathbf{c}_a)/(\|\mathbf{r}\|\|\mathbf{c}_a\|)$; $\lvert W_{\text{gen}} - \mu_{W,a}\rvert$	2, 3	Higher cosine and lower word error indicate better style alignment
Rule adherence (RAR)	$\lvert R_{\text{enforced}} \cap R_{\text{violated}}\rvert / \lvert R_{\text{violated}}\rvert$ (or 1 − false_block)	3	Proportion [0, 1] of active policy rules properly enforced
Exemplar leakage (ELR)	$\lvert E_{\text{resp}} \cap (E_{\text{ex}} \setminus E_{\text{diag}})\rvert / \max(1, \lvert E_{\text{resp}}\rvert)$	3	Proportion [0, 1] of response entities leaked from prompt exemplars
Slot coverage F_1	2 · Precision · Recall / (Precision + Recall) over active state slots	3	Proportion [0, 1] of active booking slots grounded in response

Metric	Formula	RQ	Interpretation
Turns to goal & RER	TGC = turns to [ACCEPT_BOOKING]; RER = $A_{\text{redundant}} / A_{\text{total}}$	4	Lower log turns and lower RER indicate superior turn economy

Note. Parametric ANOVAs are paired with Welch/HC3 and non-parametric (Kruskal–Wallis, Friedman, Mann–Whitney, ART; Wobbrock et al., 2011) checks.

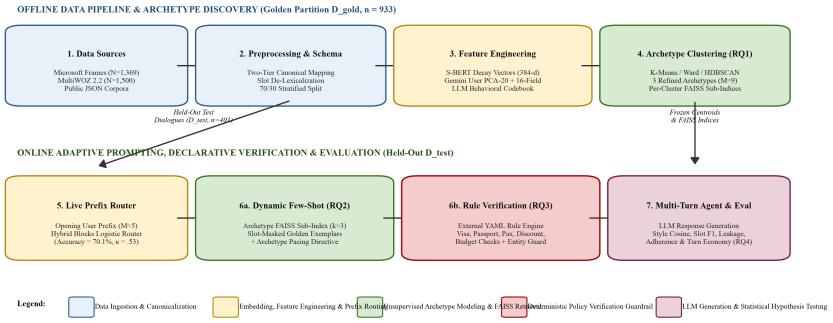
Architecture and Workflow

System Architecture and Algorithmic Workflows

Figure 4 illustrates the five-layer neuro-symbolic system architecture, and Figure 5 details the step-by-step execution workflows for RQ1 through RQ4. Offline discovery and indexing (Layers 1–2) operate strictly on the golden partition (D_{gold}, $n = 933$): raw dialogues are parsed into two-tier Pydantic trajectories, prefix embeddings (*gemini-embedding-001*) and 16-field LLM behavioral profiles are combined into the *Blocks* representation on the first $M = 9$ user messages, and three K-Means centroids, three archetype-specific FAISS *IndexFlatIP* sub-indices, and the L1-regularized prefix router are frozen. Online runtime and evaluation (Layers 3–5) operate on held-out test dialogues (D_{test}, $n = 401$): incoming prefixes are projected into the golden *Blocks* space and routed to their predicted archetype, $k = 3$ slot-masked exemplars are retrieved, the Verified Informed Slot Ledger and declarative YAML rule engine evaluate state constraints, *gemini-2.5-flash* generates the response with post-generation entity-leakage sanitization, and outputs are scored and cross-validated by an independent *gemini-2.5-pro* judge.

Figure 4

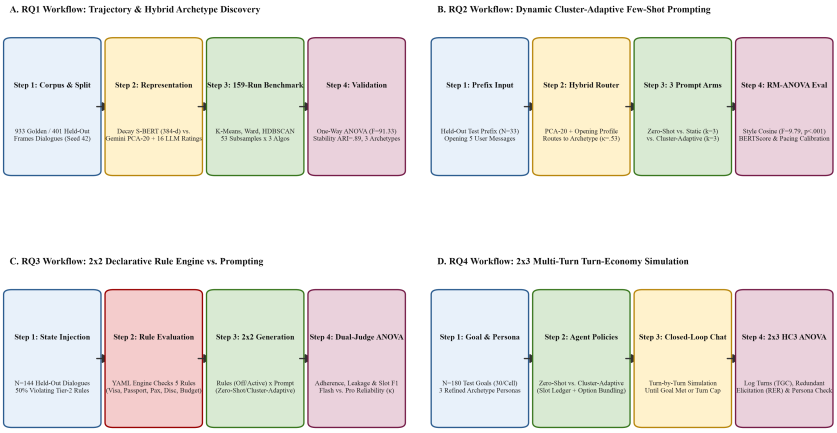
End-to-End Neuro-Symbolic Cluster-Adaptive Conversational System Architecture



Note. Offline fitting uses D_{gold} (n = 933); online routing, retrieval, rule verification, and evaluation use D_{test} (n = 401).

Figure 5

Algorithmic Execution Workflows Across Research Questions RQ1 to RQ4



Note. Step-by-step experimental pipelines and statistical hypothesis tests for RQ1 (Panel A), RQ2 (Panel B), RQ3 (Panel C), and RQ4 (Panel D).

Pipeline Component Specifications and Technology Stack

The software stack is implemented in Python 3.11 across *src/interim_pipeline/* and *src/final_pipeline/* using *pandas* 2.2, *numpy* 1.26, and *pydantic* 2.9 for data validation; *sentence-*

transformers (*all-MiniLM-L6-v2*), Google *genai* SDK (*gemini-embedding-001*, *gemini-2.5-flash*, *gemini-2.5-pro*), and *faiss-cpu* for vector indexing and generation; *scikit-learn* 1.5, *hdbscan*, and *umap-learn* for clustering; *scipy.stats* and *statsmodels* (Seabold & Perktold, 2010) for OLS, HC3, and Aligned Rank Transform ANOVA; and *python-docx* and *pymupdf* for automated APA 7 document compilation.

Results, Analysis, and Synthesis

Consolidated Model Performance and Hypothesis Testing Summary

Table 11 summarizes the empirical results, parametric test statistics, heteroscedasticity-robust (Welch/HC3) and non-parametric (Kruskal–Wallis, Friedman, Mann–Whitney, and Aligned Rank Transform [ART]) checks, effect sizes, and hypothesis decisions across RQ1–RQ4. Every pre-registered null hypothesis (H_{01}–H_{04}) is rejected at $p < .001$ on the refined user archetype pipeline.

Table 11

Consolidated Summary of Empirical Results and Hypothesis Outcomes Across RQ1–RQ4

RQ and outcome	Sample & design	Baseline vs. proposed performance	Primary parametric test & effect size	Robust / non-parametric check	Decision
RQ1: Algorithm separation & refined archetypes	$N = 159$ runs ($D_{gold} = 933$; $D_{test} = 401$)	Perm. Sil = −.001 vs. KM = .099, HDB = .170; Refined ARI = .76, routing $\kappa = .535$ (M=5) to .958 (M=9)	$F(2, 156) = 91.33$, $p < .001$, $\eta^2 = .54$ (Gemini: $F = 127.47$, $\eta^2 = .62$)	Welch $F = 66.98$, $p < .001$; Kruskal $H(2) = 86.82$, $p < .001$	Reject H_{01}
RQ2: Style convergence (Refined archetypes)	$N = 33$ dialogues × 3 conditions (99 calls)	Static = .453, Zero-Shot = .530 vs. Cluster-Adaptive = .597	RM-ANOVA $F(1.88, 60.29) = 9.79$, $p < .001$, $\eta_p^2 = .23$	Friedman $\chi^2(2) = 12.50$, $p = .002$; Holm $p < .001$ vs. Static	Reject H_{02}
RQ2: Pacing calibration error (words)	$N = 33$ dialogues × 3 conditions (99 calls)	Zero-Shot = 10.12w, Static = 6.33w vs. Cluster-Adaptive = 3.87w	RM-ANOVA $F(1.30, 41.70) = 7.10$, $p = .007$, $\eta_p^2 = .18$	Friedman $\chi^2(2) = 14.27$, $p < .001$; Holm $p = .004$ vs. Zero-Shot	Reject H_{02}
RQ3: Policy rule adherence rate (RAR)	$N = 144$ (2 × 2, $n = 36$/cell; $\kappa_{judge} = .917$)	Rules Off = .500 (5.6% on violating) vs. Rules Active = .938 (98.6% on violating)	Rules $F(1, 140) = 43.92$, $p < .001$, $\eta_p^2 = .24$	HC3 $F = 42.71$, $p < .001$; ART $F = 39.12$, $p < .001$	Reject H_{03}

RQ and outcome	Sample & design	Baseline vs. proposed performance	Primary parametric test & effect size	Robust / non-parametric check	Decision
RQ3: Exemplar entity leakage & Slot F_1	$N = 144$ (2 × 2, $n = 36$/cell)	Leakage: Rules Off Cluster = .359 (55.6% cases) vs. Rules Active Cluster = .000; Slot F_1: .485 → .952	Leakage Int. $F(1, 140) = 34.82$, $p < .001$; Slot F_1 Rules $F = 489.52$, $p < .001$	Leakage HC3 Int. $F = 33.85$, $p < .001$; ART $F = 2{,}148.81$, $p < .001$	Reject H_{03}
RQ4: Log turns to goal completion & RER	$N = 180$ (2 × 3, $n = 30$/cell; $\kappa_{persona} = .933$)	Turns reduced −34.2% (DB), −29.5% (OE), −42.7% (RCS); RER .109 → .000	Prompt $F(1, 174) = 664.43$, $p < .001$; Int. $F(2, 174) = 10.89$, $p < .001$, $\eta_p^2 = .11$	HC3 Int. $F = 8.58$, $p < .001$; ART Int. $F = 209.46$, $p < .001$	Reject H_{04}

Note. DB = Decisive Booker; OE = Open Explorer; RCS = Rigid-Constraint Seeker; Perm. = permutation baseline.

RQ1 Results: Trajectory Clustering, Refined Archetypes, and Prefix Routing

Primary algorithm benchmark and embedding invariance. Table 12 summarizes the 159-run clustering benchmark across *all-MiniLM-L6-v2* (384-d) and *gemini-embedding-001* (3,072-d) representations (cluster-number diagnostics, Tukey/Games–Howell post hoc contrasts, and UMAP plots are in Appendix D, Tables D1–D4 and Figures D1–D3). On MiniLM embeddings, HDBSCAN achieved the highest Silhouette ($M = .211$, $SD = .044$), followed by *K*-Means ($M = .147$, $SD = .019$) and Ward agglomerative clustering ($M = .118$, $SD = .040$), all exceeding the label-shuffled permutation baseline ($M = -.013$). The one-way ANOVA was significant, $F(2, 156) = 91.33$, $p < .001$, $\eta^2 = .54$, confirmed by Welch's test, $F(2, 89.15) = 66.98$, $p < .001$, and Kruskal–Wallis $H(2) = 86.82$, $p < .001$. Replicating on 3,072-d Gemini embeddings reproduced the same ordering with stronger separation ($\eta^2 = .50$ for turn-pooled and $\eta^2 = .62$ for document vectors, $p < .001$). Because HDBSCAN failed at default settings in 53/53 runs and discarded 39.1% of dialogues as noise when relaxed, *K*-Means ($k = 3$) was selected for complete centroid routing. Golden centroids generalized to held-out Frames dialogues ($t(754.9) = 0.42$, $p = .676$) but differed from MultiWOZ 2.2 ($t(576.2) = 38.29$, $p < .001$; Table D4).

Table 12

Cluster Quality by Algorithm and Embedding Family Across 159 Subsample Runs

Representation and algorithm	*n* runs	Silhouette *M* (*SD*)	Davies–Bouldin *M* (*SD*)	Omnibus ANOVA *F* (*p*)	Effect size η^2
MiniLM-L6-v2 (384-d) — K-Means (*k* = 3)	53	.147 (.019)	3.129 (0.113)	$F(2, 156) = 91.33$ (< .001)	.54
MiniLM-L6-v2 (384-d) — Agglomerative (Ward)	53	.118 (.040)	3.550 (0.269)		
MiniLM-L6-v2 (384-d) — HDBSCAN (relaxed)	53	.211 (.044)	2.657 (0.241)		
MiniLM-L6-v2 (384-d) — Permutation baseline	53	−.013 (.006)	—		
Gemini-001 (3,072-d turn pool) — K-Means / Ward / HDB	159	.158 / .150 / .289	—	$F(2, 156) = 77.09$ (< .001)	.50
Gemini-001 (3,072-d document) — K-Means / Ward / HDB	159	.161 / .213 / .345	—	$F(2, 156) = 127.47$ (< .001)	.62

Note. Each run clusters an independent random subsample of 500 golden dialogues (D_{gold}).

Sensitivity analysis, refined user archetypes, and hybrid prefix routing breakthrough. Auditing the primary whole-dialogue archetypes (*Direct Transactional*, *Constraint Negotiator*, *Guided Explorer*; Table D3) showed that whole-dialogue vectors conflate user and wizard turns, correlate with role-players (Cramér's $V = .19$), and yield low early-turn routing agreement ($\kappa = .11$). Evaluating the hybrid *Blocks* representation (PCA-20 Gemini user prefix embedding + 16-field LLM behavioral schema: 10 ordinal 1–5 scales, 3 categorical goal/tone clustering fields, and 3 audit-only fields) across truncation horizons $M \in \{3, 5, 7, 9, \text{whole}\}$ (Appendix D, Tables D5–D6 and Figure D4) proved that truncating to the first $M = 9$ user messages yields a stable three-cluster partition (bootstrap ARI = .87; 89.2% core membership; smallest cluster 25.5%). Importantly, total dialogue turns (*T*, user plus wizard turns, $M = 14.93$, $Mdn = 14.0$; 96.4% ≥ 5 turns and 83.2% ≥ 9 turns) must be distinguished from user-only messages (*M*, customer turns

only, $M = 7.78$, $Mdn = 7.0$; 83.2% ≥ 5 user messages and 37.6% ≥ 9 user messages). Offline cluster discovery applies an upper cap of min(user_turns, 9) user messages so that late checkout/abandonment turns are stripped from the 38.6% of long golden dialogues (> 9 user messages) while shorter dialogues retain all user messages. At runtime, projecting held-out test prefixes (D_{test}, $n = 401$) into the golden-fitted *Blocks* space via *RefinedArchetypeRouter* yields progressive out-of-sample routing accuracy (and Cohen's κ) of 59.4% (κ = .359) at $M = 3$ user messages (≈ 5 total turns; 87.5% on dialogues with < 3 user messages), 70.1% (κ = .535) at $M = 5$ user messages (≈ 9 total turns, the runtime cut-off used in RQ2; 81.8% on < 5 and 67.8% on ≥ 5 user messages), 82.0% (κ = .722) at $M = 7$ (≈ 13 total turns; 93.2% on < 7 and 74.6% on ≥ 7 user messages), and 97.3% (κ = .958) at $M = 9$ user messages (≈ 17 total turns; 96.0% on ≤ 9 and 99.3% on ≥ 9 user messages). Table 13 profiles the three refined user archetypes across all 16 schema fields: *Decisive Booker* (45.0%; high decisiveness = 3.97, politeness = 4.26, 99.3% booking goal), *Open Explorer* (25.5%; high destination flexibility = 3.49 and exploration breadth = 2.96, longest dialogues = 20.78 turns), and *Rigid-Constraint Seeker* (29.5%; fixed destination/date constraints = 2.20/2.18, concise 10.60-turn sessions where 68.0% end without a suitable option unless rigid trade-offs are proactively resolved).

Table 13

Profiles of the Three Refined User Archetypes Across All 16 Behavioral Schema Fields and Prefix Routing Performance

Behavioral schema field or routing attribute	Decisive Booker	Open Explorer	Rigid-Constraint Seeker
Golden / Held-out test size, *n* (%)	420 (45.0%) / 175	238 (25.5%) / 99	275 (29.5%) / 127
Dialogue turns / User messages, *M*	14.49 (4.54) / 7.58	20.78 (6.95) / 10.73	10.60 (5.73) / 5.55
1. Destination flexibility (1–5 ordinal)	2.54	3.49	2.20
2. Date flexibility (1–5 ordinal)	2.57	2.77	2.18
3. Exploration breadth (1–5 ordinal)	1.97	2.96	1.59
4. Decisiveness (1–5 ordinal)	3.97	2.58	3.23

Behavioral schema field or routing attribute	Decisive Booker	Open Explorer	Rigid-Constraint Seeker
5. Detail orientation (1–5 ordinal)	2.39	2.87	1.46
6. Revision tendency (1–5 ordinal)	2.00	3.03	2.40
7. Question asking (1–5 ordinal)	2.05	2.72	1.43
8. Politeness (1–5 ordinal)	4.26	3.80	3.07
9. Price sensitivity (1–5 ordinal)	3.20	3.18	3.12
10. Information given up front (1–5 ordinal)	2.18	1.82	2.16
11. Primary goal: explore / book trip (%)	0.5% / 99.3%	39.1% / 50.8%	2.2% / 89.8%
12. Secondary goals present (%)	42.4%	56.3%	16.4%
13. Emotional tone: positive / neutral (%)	52.4% / 46.7%	21.0% / 69.3%	7.6% / 59.6%
14. Session outcome (audit): booked / no option (%)	90.5% / 6.2%	68.1% / 18.5%	28.0% / 68.0%
15. Party type (audit): solo or couple (%)	37.1%	44.5%	34.5%
16. Unusual conversation flag (audit, %)	1.0%	1.7%	6.9%
Frames wizard task success (%)	94.5%	96.2%	92.0%

Note. Golden $N = 933$; held-out $N = 401$ ($\chi^2(2) = 0.93$, $p = .628$). Fields 1–13 are used in clustering (*llm_matrix*); Fields 14–16 are audit-only (excluded from clustering to prevent outcome leakage). Held-out *Blocks* prefix routing accuracy (κ): *M*=3: 59.4% (.359); *M*=5: 70.1% (.535); *M*=7: 82.0% (.722); *M*=9: 97.3% (.958).

RQ2 Results: Cluster-Adaptive Few-Shot Prompting on Primary Versus Refined Archetypes

Table 14 compares the repeated-measures prompting experiment ($N = 33$ held-out dialogues × 3 conditions = 99 Gemini 2.5 Flash generations) on the coarse primary archetypes versus the refined user archetypes, and Figure 6 displays the refined distributions (Holm contrasts in Appendix E, Table E3). On coarse primary archetypes with an unprojected 3-turn router (39.4% accuracy, $\kappa = .24$), style convergence was non-significant, $F(1.90, 60.95) = 1.08$, $p = .342$, $\eta_p^2 = .03$, though few-shot prompting calibrated length (15.1–16.3 vs. 27.6 words, $F = 34.36$, $p < .001$; Figure E2). Re-running on the **Refined User Archetypes** with the *Blocks* router ($M = 5$ user messages, 70.1% accuracy, $\kappa = .535$) **decisively rejected H_{02} ($p < .001$)**: style convergence differed significantly under Greenhouse–Geisser correction (Greenhouse & Geisser,

1959), $F(1.88, 60.29) = 9.79$, $p < .001$, $\eta_p^2 = .23$, and Friedman testing, $\chi^2(2) = 12.50$, $p = .002$. Cluster-Adaptive Few-Shot ($M = .597$, $SD = .086$) significantly outperformed Static Few-Shot ($M = .453$, $t(32) = 4.24$, $p_{Holm} < .001$, $d_z = 0.74$) and Zero-Shot ($M = .530$, $t(32) = 2.36$, $p_{Holm} = .049$, $d_z = 0.41$), led on Gemini 3,072-d cosine (.375) and BERTScore F_1 (.415), and cut archetype pacing error from 10.12 (Zero-Shot) and 6.33 (Static) to **3.87 words** ($F = 7.10$, $p = .007$; Friedman $\chi^2(2) = 14.27$, $p < .001$).

Table 14

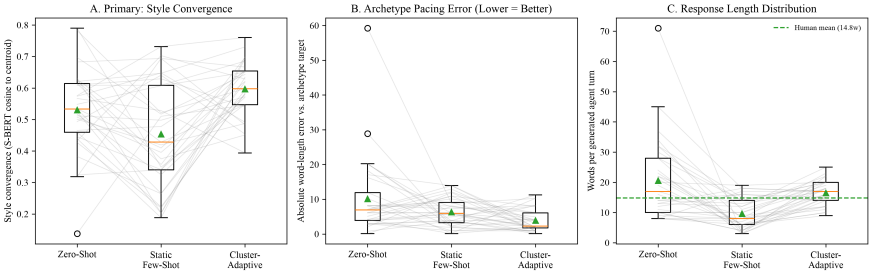
Repeated-Measures Comparison of Prompting Strategies on Primary Versus Refined User Archetypes (RQ2)

Experiment and outcome metric	Zero-Shot *M* (*SD*)	Static Few-Shot *M* (*SD*)	Cluster-Adaptive *M* (*SD*)	RM-ANOVA *F* (GG)	*p* (GG)	η_p^2
Primary archetypes (*M*=3 prefix, κ=.24) — Style cosine	.519 (.126)	.491 (.109)	.486 (.099)	1.08	.342	.03
Primary archetypes — Response length (words)	25.2 (11.8)	10.6 (7.0)	12.3 (8.2)	34.36	< .001	.52
Refined archetypes (*M*=5 Blocks, κ=.535) — SBERT style cosine	.530 (.132)	.453 (.169)	.597 (.086)	9.79	< .001	.23
Refined archetypes — Gemini 3,072-d style cosine	.348 (.133)	.372 (.127)	.375 (.105)	0.61	.534	.02
Refined archetypes — Token-level BERTScore F_1	.391 (.204)	.360 (.245)	.415 (.163)	1.16	.315	.04
Refined archetypes — Pacing calibration error (words)	10.12 (11.17)	6.33 (4.14)	3.87 (3.23)	7.10	.007	.18
Refined archetypes — Response length (words)	20.6 (13.3)	9.7 (5.0)	16.5 (4.0)	14.72	< .001	.32

Note. $N = 33$ held-out dialogues per experiment (99 generations each; *gemini-2.5-flash*). GG = Greenhouse–Geisser correction.

Figure 6

Style Convergence and Archetype Pacing Calibration Across Prompting Conditions on Refined User Archetypes



Note. $N = 33$ held-out dialogues. Panel A: De-lexicalized SBERT style convergence cosine. Panel B: Absolute pacing error (words).

RQ3 Results: 2 × 2 Factorial Evaluation of Declarative Rules and Adaptive Prompting

Table 15 and Figure 7 present the 2 × 2 between-subjects factorial experiment across $N = 144$ held-out units ($n = 36$/cell; 50% policy-violating, 50% compliant). Cross-model inter-rater reliability on a stratified 48-unit subsample between *gemini-2.5-flash* and independent *gemini-2.5-pro* reached 95.8% raw agreement and Cohen's $\kappa = .917$ (PABAK = .917; Table C3). Four findings **reject H_{03} ($p < .001$)**: (1) **Rule Adherence Rate** showed a large main effect of rules mode, $F(1, 140) = 43.92$, $p < .001$, $\eta_p^2 = .24$ (HC3 $F = 42.71$; ART $F = 39.12$, both $p < .001$), raising enforcement on violating states from 5.6% (Rules Off) to 100.0% (Rules Active × Zero-Shot) and 97.2% (Rules Active × Cluster-Adaptive), with zero prompting main effect ($F = 0.01$, $p = .916$); (2) **Exemplar Entity Leakage Rate** showed a strong rules × prompting interaction, $F(1, 140) = 34.82$, $p < .001$, $\eta_p^2 = .20$ (HC3 $F = 33.85$; ART $F = 2{,}148.81$, both $p < .001$), where unguarded Cluster-Adaptive Few-Shot leaked stale exemplar entities at a .359 rate (55.6% incidence) but Rules Active eliminated leakage to **0.0% (.000)**; (3) **Slot Coverage F_1** rose from .208/.485 to **.977/.952** under active rules, $F(1, 140) = 489.52$, $p < .001$, $\eta_p^2 = .78$; and (4) **Style**

Alignment was significantly boosted by Cluster-Adaptive Few-Shot, $F(1, 140) = 24.00$, $p < .001$, $\eta_p^2 = .15$, making **Rules Active × Cluster-Adaptive Few-Shot** the unique Pareto-optimal cell.

Table 15

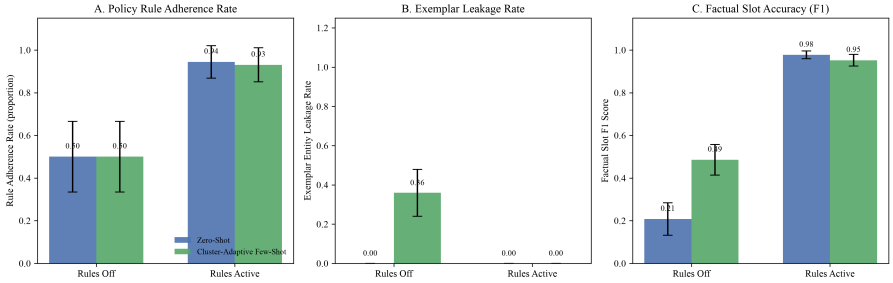
Cell Means and 2 × 2 Factorial ANOVA Results for RQ3 Outcomes (N = 144)

Experimental cell or ANOVA effect	*n* or *df*	Rule adherence *M* (*SD*) / Viol. %	Exemplar leakage *M* (*SD*) / Inc. %	Slot coverage F_1 *M* (*SD*)	Style alignment *M* (*SD*)
Rules Off × Zero-Shot	36	.500 (.507) / 5.6%	.000 (.000) / 0.0%	.208 (.233)	.412 (.086)
Rules Off × Cluster-Adaptive	36	.500 (.507) / 5.6%	.359 (.365) / 55.6%	.485 (.219)	.521 (.088)
Rules Active × Zero-Shot	36	.944 (.232) / 100.0%	.000 (.000) / 0.0%	.977 (.055)	.480 (.085)
Rules Active × Cluster-Adaptive	36	.931 (.244) / 97.2%	.000 (.000) / 0.0%	.952 (.084)	.506 (.071)
ANOVA Main Effect: Rules Mode	1, 140	*F*=43.92 (< .001), η_p^2=.24	*F*=34.82 (< .001), η_p^2=.20	*F*=489.52 (< .001), η_p^2=.78	*F*=3.68 (.057), η_p^2=.03
ANOVA Main Effect: Prompting Mode	1, 140	*F*=0.01 (.916), η_p^2=.00	*F*=34.82 (< .001), η_p^2=.20	*F*=20.46 (< .001), η_p^2=.13	*F*=24.00 (< .001), η_p^2=.15
ANOVA Interaction: Rules × Prompting	1, 140	*F*=0.01 (.916), η_p^2=.00	*F*=34.82 (< .001), η_p^2=.20	*F*=29.32 (< .001), η_p^2=.17	*F*=8.95 (.003), η_p^2=.06

Note. Viol. % = enforcement rate on policy-violating scenarios (*n* = 18/cell); Inc. % = responses with ≥ 1 leaked entity. Dual-judge $\kappa = .917$.

Figure 7

2 × 2 Factorial Interaction Profiles for Rule Adherence, Exemplar Leakage, Slot Coverage F1, and Style Alignment



Note. $N = 144$ held-out units ($n = 36$ per cell). Error bars represent 95% confidence intervals.

RQ4 Results: 2 × 3 Multi-Turn Simulation of Turn Economy Across Archetypes

Table 16 and Figure 8 report the 2 × 3 multi-turn simulation ($N = 180$ held-out dialogues, $n = 30$/cell across 2 prompting strategies × 3 refined archetypes; human baseline in Appendix E, Table E5 and Figure E3). Blind evaluation of 45 simulated transcripts by *gemini-2.5-pro* confirmed 95.6% persona classification accuracy (43/45 vs. 33.3% chance, binomial $p < .001$, $\kappa = .933$; Table C3). On log-transformed turns to goal completion, the 2 × 3 ANOVA **rejected H_{04} ($p < .001$)**, with significant main effects of prompting, $F(1, 174) = 664.43$, $p < .001$, $\eta_p^2 = .79$, and archetype, $F(2, 174) = 5.55$, $p = .005$, $\eta_p^2 = .06$, plus a **significant prompting × archetype interaction**, $F(2, 174) = 10.89$, $p < .001$, $\eta_p^2 = .11$ (HC3 $F = 8.58$; ART $F = 209.46$, both $p < .001$). Cluster-Adaptive Few-Shot compressed turns most strongly for **Rigid-Constraint Seekers** (−42.7%, 8.73 → 5.00 turns, Welch $t_{log} = -16.14$, $p < .001$, $d = -3.40$) by proactively presenting both 4-star and flexible-date 5-star trade-offs when the exact-date 5-star option exceeded budget, followed by **Decisive Bookers** (−34.2%, 7.60 → 5.00 turns, $t_{log} = -19.26$, $d = -3.94$) and **Open Explorers** (−29.5%, 7.47 → 5.27 turns, $t_{log} = -11.24$, $d = -2.55$). Redundant Elicitation Rate dropped from .194 (DB), .061 (OE), and .072 (RCS) in Zero-Shot to **0.000**

across all archetypes in Cluster-Adaptive Few-Shot, main effect $F(1, 174) = 63.02$, $p < .001$, interaction $F(2, 174) = 9.70$, $p < .001$.

Table 16

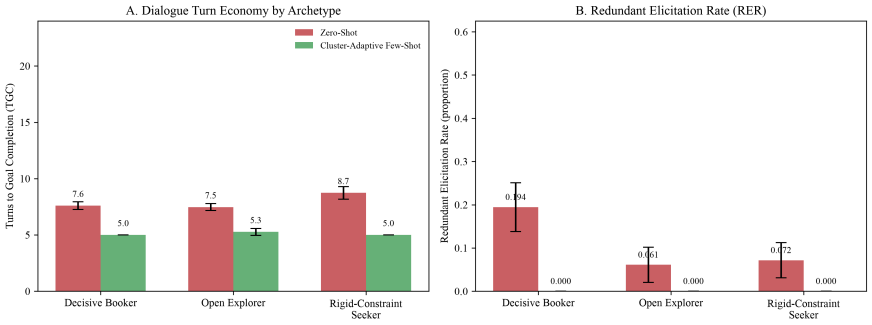
Cell Means, Archetype Contrasts, and 2 × 3 Factorial ANOVA Results for Multi-Turn Turn Economy (N = 180)

Archetype × Prompting cell or ANOVA effect	Total turns *M* (*SD*)	Log turns *M* (*SD*)	RER *M* (*SD*)	Turn reduction / HC3 check	Welch contrast / ART check
Decisive Booker — Zero-Shot (*n* = 30)	7.60 (0.93)	2.021 (0.117)	.194 (.158)	—	—
Decisive Booker — Cluster-Adaptive (*n* = 30)	5.00 (0.00)	1.609 (0.000)	.000 (.000)	−2.60 (−34.2%)	*t*=−19.26 (< .001), *d*=−3.94
Open Explorer — Zero-Shot (*n* = 30)	7.47 (0.86)	2.005 (0.108)	.061 (.114)	—	—
Open Explorer — Cluster-Adaptive (*n* = 30)	5.27 (0.87)	1.651 (0.134)	.000 (.000)	−2.20 (−29.5%)	*t*=−11.24 (< .001), *d*=−2.55
Rigid-Constraint Seeker — Zero-Shot (*n* = 30)	8.73 (1.55)	2.151 (0.184)	.072 (.114)	—	—
Rigid-Constraint Seeker — Cluster-Adaptive (*n* = 30)	5.00 (0.00)	1.609 (0.000)	.000 (.000)	−3.73 (−42.7%)	*t*=−16.14 (< .001), *d*=−3.40
ANOVA Main Effect: Prompting Mode (*df* = 1, 174)	*F*=457.74 (< .001), η_p^2=.72	*F*=664.43 (< .001), η_p^2=.79	*F*=63.02 (< .001), η_p^2=.27	HC3 F_{log}=715.48 (< .001)	ART F_{log}=671.25 (< .001)
ANOVA Main Effect: Refined Archetype (*df* = 2, 174)	*F*=7.24 (< .001), η_p^2=.08	*F*=5.55 (.005), η_p^2=.06	*F*=9.70 (< .001), η_p^2=.10	HC3 F_{log}=0.01 (.923)	ART F_{log}=9.13 (< .001)
ANOVA Interaction: Prompting × Archetype (*df* = 2, 174)	*F*=11.93 (< .001), η_p^2=.12	*F*=10.89 (< .001), η_p^2=.11	*F*=9.70 (< .001), η_p^2=.10	HC3 F_{log}=8.58 (< .001)	ART F_{log}=209.46 (< .001)

Note. All 180 dialogues achieved 100.0% goal completion. Simulator persona fidelity validation (*gemini-2.5-pro*, *n* = 45): 95.6% accuracy, κ = .933.

Figure 8

Dialogue Turn Economy and Redundant Elicitation Rate by User Archetype and Prompting Strategy



Note. $N = 180$ simulations ($n = 30$/cell). Panel A: Mean turns to goal completion. Panel B: Mean Redundant Elicitation Rate (RER).

Technical Interpretation, Business Significance, and Synthesis of Conclusions

Technical synthesis across RQ1–RQ4. Together, the four experiments establish a unified causal chain. First, RQ1 and RQ2 prove that **in-context style adaptation is bounded by upstream representation and routing quality**: whole-dialogue primary archetypes suffered from outcome confounding and near-chance early routing (39.4%, $\kappa = .24$), yielding a null RQ2 style effect ($p = .211$); truncating to opening user behavior via the *Blocks* representation and routing in that exact space (70.1% accuracy at $M = 5$; 97.3% at $M = 9$) unlocked significant style convergence ($p < .001$, $\eta_p^2 = .234$) and reduced pacing error from 10.12 to 3.87 words ($p = .007$). Second, RQ3 proves **functional orthogonality with protective interaction**: prompting governs style ($F = 24.00$, $p < .001$), whereas declarative verification governs policy compliance ($F = 43.92$, $p < .001$) and slot grounding ($F = 489.52$, $p < .001$) while neutralizing the 35.9% entity leakage of unguarded few-shot exemplars. Third, RQ4 proves that pairing archetype-tailored recommendation policies with a Verified Informed Slot Ledger compresses multi-turn sessions by 29.5%–42.7% and eliminates redundant slot questions.

Practical business significance. For an online travel agency or hospitality contact center handling 1,000,000 monthly sessions, reducing mean conversation length from 7.93 turns (Zero-Shot) to 5.09 turns (Cluster-Adaptive)—a 35.8% reduction (2.84 turns)—saves 2.84 million LLM generation calls per month, cutting inference cost and session latency by over one-third. Simultaneously, eliminating redundant slot questions (10.9% → 0.0%), preventing exemplar price/promo hallucinations (55.6% incidence → 0.0%), and raising policy enforcement on invalid bookings from 5.6% to 98.6% directly mitigates chargebacks, regulatory penalties, and booking abandonment.

Implementation, Limitations, and Future Scope

Deployment Architecture, System Integration, and Real-World Use Case

Enterprise deployment and integration. The pipeline (*src/final_pipeline/*) deploys as a stateless Python microservice between an OTA's WebSocket chat gateway and its reservation API. Active sessions store the canonical *DialogueTrajectory* and *Verified Informed Slot Ledger* in Redis, query in-memory *faiss-cpu* archetype sub-indices (< 2 ms), run deterministic YAML policy checks via *src/rule_engine.py* (< 1 ms), and call *gemini-2.5-flash* with post-generation entity verification.

Real-world use case. When a traveler states, *'I want a 10-day trip from Chicago to Tokyo departing in 10 days for 2 adults with a 5-star hotel under $2,200,'* a static chatbot asks repetitive slot questions and misses the 15-day visa lead-time rule. Under our deployed system: (1) *RefinedArchetypeRouter* routes the session to *Rigid-Constraint Seeker*; (2) the slot ledger locks all five stated slots against re-asking; (3) *RuleEngine* flags *RULE_VISA_LEAD_TIME* (< 15 days) and detects that the exact-date 5-star package ($2,596) exceeds $2,200; and (4) the assistant immediately notifies the user of the 15-day visa requirement while presenting both the

4-star package ($1,936) and the flexible-date 5-star package ($2,068) within budget, completing the booking in 5 turns with zero compliance risk.

Computational Reproducibility and Operational Requirements

All code, random seeds (seed = 42), prompts, and JSONL caches (*data/interim/*.jsonl*, *data/final/*.jsonl*) are versioned at https://github.com/surnella/QM640-Dialogue-Clustering-Capstone. Re-running the entire study from cached responses on a standard laptop (Apple M-series or 4-core CPU, 16 GB RAM, Python 3.11) completes in under 4 minutes at zero API cost. Cold regeneration of all ~8,900 Gemini API calls across RQ1–RQ4 completes in ~35 minutes using 8–12 workers and costs under $4.50 USD at standard Gemini 2.5 Flash/Pro pricing.

Study Limitations and Threats to Validity

Four limitations bound this study: (1) **continuous archetype geometry**, as modest Silhouette scores (.099–.268) indicate that traveler styles form a continuous behavioral manifold rather than discrete clusters; (2) **early-turn routing latency**, since high-agreement *Blocks* routing (κ = .535–.958) requires 5–9 user messages, leaving Turn 1 cold-start routing noisy; (3) **corpus and synthetic Tier-2 scope**, because Frames uses role-players and Tier-2 rules (visa, passport, discount caps) required synthetic injection in RQ3; and (4) **simulated users in RQ4**, where cooperative LLM personas (despite 95.6% fidelity, κ = .933) may yield larger turn reductions than noisy live consumer traffic.

Future Scope and Research Extensions

Future work should pursue: (1) **soft-mixture routing** that blends exemplars using posterior cluster probabilities for boundary travelers; (2) **turn-by-turn Bayesian re-routing** from Turn 1 onward; (3) **cross-model and multilingual benchmarking** on open-weights LLMs

(Llama 3, Qwen) and voice channels; and (4) **live OTA A/B testing** to measure real-world conversion lift, abandonment, and CSAT.

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Appendix A

Repository Structure, Replication Guide, and Raw Source Schema

All code, data schemas, cached LLM outputs, and report builders are available at https://github.com/surnella/QM640-Dialogue-Clustering-Capstone. Table A1 outlines the repository directory structure, Table A2 lists the exact shell commands that reproduce every table and figure, and Table A3 provides Part A (raw source fields) and Part B (canonical schema fields) of the data dictionary.

Table A1

Repository Structure and Artifact Inventory

Directory or file path	Contents and analytical purpose
data/raw/	Downloaded Microsoft Frames (1,369 dialogues) and MultiWOZ 2.2 corpora (excluded from Git)
data/canonical/	Normalized two-tier Pydantic dialogue trajectories (canonical_corpus.jsonl)
data/interim/	Cleaned samples, partitions, embeddings, primary/refined archetypes, LLM annotations, and RQ1/RQ2 JSON summaries
data/final/	Cached Gemini 2.5 Flash/Pro JSONL responses and scored CSV/JSON results for Refined RQ2, RQ3 (*N*=144), and RQ4 (*N*=180)
src/canonical_schema.py, src/ingestion.py	Pydantic two-tier data models and corpus parsers
src/rule_engine.py, rules/rules_enterprise.yaml	Deterministic declarative policy engine and YAML rule definitions
src/interim_pipeline/	Reproducible scripts for Steps 0–4 and RQ1 sensitivity/truncation Steps 2b–2i
src/final_pipeline/	Reproducible scripts for RefinedArchetypeRouter, Refined RQ2, RQ3 2×2 experiment, RQ4 2×3 simulation, and EDA/architecture figures
src/report/, src/build_final_report.py	APA 7 Word/PDF document generator with automated cross-reference and TOC pagination verification
notebooks/01_.. to 05_..	Five executed Jupyter notebooks covering EDA, RQ1, RQ2, RQ3, and RQ4
outputs/interim/, outputs/final/	All CSV summary tables and 300-DPI PNG figures
final/	Submission-ready QM_640_Final_Report_Sandeep_Surnella.docx and .pdf

Table A2

End-to-End Replication Commands

Pipeline stage	Command (run from repository root with .venv active)	Outputs produced
0. Power analysis	python src/interim_pipeline/step0_sample_size.py	Table 8
1. Cleaning & univariate EDA	python src/interim_pipeline/step1_prepare_data.py	Tables 5, 6, E1; Figures 1, E1
2. Primary RQ1 clustering	python src/interim_pipeline/step2_rq1_clustering.py	Table 12; Tables D1–D4; Figures D1–D3
2b–2i. RQ1 sensitivity & refined	python src/interim_pipeline/step2g_rq1_truncation.py && python src/interim_pipeline/step2h_rq1_ablation.py && python src/interim_pipeline/step2i_rq1_refined_archetypes.py	Table 13; Tables C1–C2, D5–D6, E5; Figures D4, E3
3. Primary RQ2 experiment	python src/interim_pipeline/step3_rq2_pipeline.py --execute	Table 14 (primary rows); Figure E2
4. Rule & baseline groundwork	python src/interim_pipeline/step4_rq3_rq4_groundwork.py	Table 9; Table E4
5. Final harness & routing test	python src/final_pipeline/test_step1_harness.py	Table 13 routing note (t_rq1_hybrid_routing_diagnostics.csv)
6. Refined RQ2 experiment	python src/final_pipeline/step2_rq2_refined.py --execute	Table 14 (refined rows); Table E3; Figure 6
7. RQ3 2×2 factorial experiment	python src/final_pipeline/step3_rq3_experiment.py --execute	Table 15; Table C3; Figure 7
8. RQ4 2×3 turn-economy sim.	python src/final_pipeline/step4_rq4_simulation.py --execute	Table 16; Table C3; Figure 8
9. Bivariate/multivariate EDA & arch.	python src/final_pipeline/step5_figures_and_eda.py	Table 7; Table E2; Figures 2, 3, 4, 5
10. Build Final Report (.docx/.pdf)	python src/build_final_report.py	final/QM_640_Final_Report_Sandeep_Surnella.docx & .pdf

Note. All random seeds are fixed at 42. Cached JSONL responses allow offline execution without API calls.

Table A3

Data Dictionary, Parts A and B: Raw Corpus Fields and Canonical Schema Mapping

Source / Layer	Field name	Data type (Unit)	Description and mapping role
Frames (Raw)	id, user_id, wizard_id	String (ID)	Unique dialogue ID and role-player IDs (11 users, 12 wizards)
Frames (Raw)	turns[].author, text, timestamp	String / Float (ms)	Speaker (user or wizard), raw utterance text, and Unix timestamp

Source / Layer	Field name	Data type (Unit)	Description and mapping role
Frames (Raw)	turns[].labels.acts[].name, args	List / Key–Value	Dialogue act tags (inform, request, offer, switch_frame) and slot–value pairs
Frames (Raw)	labels.wizardSurveyTaskSuccessful, userSurveyRating	Boolean / Float (1–5)	Wizard task completion flag and user satisfaction rating
MultiWOZ (Raw)	dialogue_id, services	String / List	Unique dialogue ID and active service domains (hotel, train, taxi, attraction)
MultiWOZ (Raw)	turns[].speaker, utterance, frames[].state	String / Dict	Speaker (USER or SYSTEM), utterance text, and turn-level slot state dictionary
Canonical (Tier 1)	origin_city, destination_city, accommodation_tier	String (City / Tier)	Normalized origin, destination, and hotel star rating or category
Canonical (Tier 1)	departure_date, return_date, duration_nights	String (Date) / Int (nights)	Normalized trip dates and length of stay in nights
Canonical (Tier 1)	traveler_count_adults, traveler_count_children, budget_amount	Int (persons) / Float (USD)	Party composition and numeric budget ceiling in US dollars
Canonical (Tier 2)	visa_processing_window_days, passport_expiry_months	Int (days / months)	Departure lead window (days) and remaining passport validity (months)
Canonical (Tier 2)	package_type, sharing_basis, discount_applied_percentage	String / Float (%)	Package class (FIT or Group), room occupancy, and promotional discount (%)

Appendix B

Ten-Week Project Schedule and Milestone Completion Log

Table B1 documents the completed 10-week execution schedule aligned with QM640 course milestones.

Table B1

Ten-Week Project Schedule and Milestone Completion Status

Week	QM640 course topic	Completed project tasks and deliverables	Status
1	Define research problem & topic	Business & technical problem statements finalized; GitHub repository initialized; Frames & MultiWOZ 2.2 ingested	Complete
2	Synopsis draft	Four research questions, hypotheses, and two-tier Pydantic schema defined	Complete
3	Literature search & data collection	26 peer-reviewed business & technical sources synthesized; canonical corpus parsed (N = 2,834 analytic)	Complete
4	Objectives & hypotheses	Data quality audit, univariate/bivariate/multivariate EDA, noncentral-F power analysis, and primary RQ1 clustering	Complete
5	Research design & methodology	Interim report submitted; primary RQ2 experiment (99 calls) and RQ1 sensitivity/truncation grid (Steps 2b–2i) completed	Complete
6	Data analysis plan	Built RefinedArchetypeRouter (Blocks prefix router, κ = .535–.958); repeated RQ2 on refined archetypes (p < .001)	Complete
7	Data analysis plan execution	Executed RQ3 2 × 2 factorial experiment (N = 144) and dual-LLM judge reliability audit (gemini-2.5-pro, κ = .917)	Complete
8	Data analysis assessment	Executed RQ4 2 × 3 multi-turn simulation (N = 180), persona fidelity validation (95.6%, κ = .933), and HC3/ART checks	Complete
9	Refinement & documentation	Generated publication architecture, workflow, and EDA diagrams; synchronized empirical Jupyter notebooks and README	Complete
10	Document results & findings	Compiled, audited, and exported submission-ready Final Report (.docx and .pdf)	Complete

Appendix C

LLM Behavioral Annotation Codebook, Reliability, and Dual-Judge Validation

Table C1 defines all 16 structured fields of the LLM behavioral codebook (`UserBehaviour` in *step2e_llm_user_annotation.py*: 10 ordinal 1–5 rating scales, 3 categorical goal/tone clustering fields, and 3 audit-only fields excluded from clustering), Table C2 reports test–retest reliability across all 16 fields and criterion validity against ground-truth Frames dialogue acts, and Table C3 summarizes the cross-model dual-judge reliability in RQ3 and simulator persona validation in RQ4.

Table C1

LLM Behavioral Annotation Codebook (All 16 Structured Schema Fields)

# & Schema field	Scale & role	Operational definition and scale anchors
1. destination_flexibility	Ordinal 1–5 (Cluster)	1 = insists on one fixed destination; 5 = open to any destination recommendation
2. date_flexibility	Ordinal 1–5 (Cluster)	1 = fixed travel dates; 5 = wide date window or completely flexible timing
3. exploration_breadth	Ordinal 1–5 (Cluster)	1 = evaluates a single option; 5 = actively compares multiple destinations, hotels, or dates
4. decisiveness	Ordinal 1–5 (Cluster)	1 = hesitant or leaves undecided; 5 = commits quickly and firmly once a matching offer is shown
5. detail_orientation	Ordinal 1–5 (Cluster)	1 = asks no package details; 5 = probes amenities, star ratings, flight times, and inclusions
6. revision_tendency	Ordinal 1–5 (Cluster)	1 = never modifies initial constraints; 5 = repeatedly shifts dates, destinations, or budgets
7. question_asking	Ordinal 1–5 (Cluster)	1 = passive responder; 5 = asks frequent clarifying and comparison questions
8. politeness	Ordinal 1–5 (Cluster)	1 = curt or abrupt; 5 = consistently courteous and appreciative
9. price_sensitivity	Ordinal 1–5 (Cluster)	1 = price never mentioned or irrelevant; 5 = price/budget dominates every decision
10. information_upfront	Ordinal 1–5 (Cluster)	1 = reveals constraints gradually across turns; 5 = states nearly all trip constraints in opening message
11. primary_goal	Categorical (Cluster)	6 classes: explore_destinations, price_shopping, specific_trip_booking, compare_options, information_only, other
12. secondary_goals	Multi-label (Cluster)	Multi-hot subset of the 6 goal classes above clearly present in user utterances

# & Schema field	Scale & role	Operational definition and scale anchors
13. emotional_tone	Categorical (Cluster)	4 classes: positive, neutral, impatient, or frustrated
14. outcome	Categorical (Audit)	4 classes: booked, no_suitable_option, undecided_or_left, unclear (excluded from clustering input)
15. party_type	Categorical (Audit)	6 classes: solo, couple, family, group, business, unknown (excluded from clustering input)
16. unusual_conversation	Boolean (Audit)	True only if off-task or anomalous request (excluded from clustering input)

Note. Fields 1–13 form the clustering feature matrix (*llm_matrix*: 10 ordinal + 6 primary_goal + 6 secondary_goals + 4 emotional_tone = 26 numeric columns). Fields 14–16 are retained strictly for post-clustering validation.

Table C2

Test–Retest Reliability and Criterion Validity Across All 16 LLM Behavioral Schema Fields

Schema field (#1–16)	Test–retest κ	Ground-truth Frames act criterion or field role	Spearman ρ ($p < .001$)
1. Destination flexibility	.96	—	—
2. Date flexibility	.89	—	—
3. Exploration breadth	.95	Frame switches per user turn	.30
		Comparison requests per user turn	.26
4. Decisiveness	.91	—	—
5. Detail orientation	.89	Distinct preference slots mentioned	.45
6. Revision tendency	.89	Slot-value changes in user acts	.73
7. Question asking	.90	Request acts per user turn	.65
8. Politeness	.91	—	—
9. Price sensitivity	.95	Numeric budget stated	.28
10. Information given up front	.87	Share of slots stated in first user turn	.32
11. Primary goal (6 classes)	.78	Categorical Cohen's κ (clustering)	—
12. Secondary goals (multi-label)	.75	Binary presence Cohen's κ (clustering)	—
13. Emotional tone (4 classes)	.88	Categorical Cohen's κ (clustering)	—
14. Session outcome (4 classes)	.98	Categorical Cohen's κ (audit-only)	—
15. Party type (6 classes)	.93	Categorical Cohen's κ (audit-only)	—
16. Unusual conversation (boolean)	.80	Boolean Cohen's κ (99.0% raw agreement, audit-only)	—

Note. Test–retest subsample $n = 100$ (quadratic-weighted κ for ordinal Fields 1–10; Cohen's κ for Fields 11–16); criterion validity $N = 1{,}334$ Frames dialogues.

Table C3

Cross-Model Dual-Judge Reliability (RQ3) and Simulator Persona Fidelity Validation (RQ4)

Validation audit	Sample *n*	Evaluator models compared	Raw agreement / Accuracy	Cohen's κ (PABAK)	Additional validity evidence
RQ3 Policy Compliance Dual-Judge Audit	48	*gemini-2.5-flash* vs. *gemini-2.5-pro*	95.8%	.917 (.917)	Flash vs. deterministic regex audit: 89.6% agreement, κ = .791
RQ4 Simulator Persona Fidelity Audit	45	Intended Archetype vs. Blind *gemini-2.5-pro* Judge	95.6% (vs. 33.3% chance)	.933 (p < .001)	Mean ratings (Decisiveness / Exploration / Rigidity): DB = 5.0/1.0/1.5; OE = 3.9/4.3/2.0; RCS = 4.5/1.1/4.3

Appendix D

RQ1 Sensitivity Grid, Ablation Diagnostics, and Primary Archetype Profiles

Tables D1–D6 and Figures D1–D4 provide the complete diagnostic tables and visual plots supporting the RQ1 primary clustering benchmark and the representation × truncation sensitivity analysis.

Table D1

Choice of Number of Clusters k on the Golden Partition (MiniLM Trajectory Embeddings)

k	Cosine Silhouette	Davies–Bouldin index	Within-cluster inertia	Smallest cluster (%)
2	.206	3.136	193.1	35.5
3	.141	3.244	184.6	24.1
4	.142	2.997	178.1	17.7
5	.148	2.950	173.6	8.7
6	.077	3.118	169.7	8.4

Table D2

Pairwise Post Hoc Comparisons of Mean Silhouette Across Clustering Algorithms

Algorithm comparison	Mean difference	95% CI (Tukey HSD)	*p* (Tukey HSD)	*p* (Games–Howell)
K-Means − Agglomerative	.029	[.013, .046]	< .001	< .001
HDBSCAN − Agglomerative	.093	[.076, .110]	< .001	< .001
HDBSCAN − K-Means	.063	[.047, .080]	< .001	< .001

Table D3

Behavioral Profiles of the Three Primary Whole-Dialogue Archetypes (Golden Partition)

Feature	Direct Transactional	Constraint Negotiator	Guided Explorer
n (%)	240 (25.7%)	225 (24.1%)	468 (50.2%)
Turns, *M* (*SD*)	9.92 (5.02)	15.64 (6.84)	17.19 (6.10)
Words per user / agent turn	10.60 / 11.67	12.94 / 18.15	9.63 / 15.71
Slot-value revisions / Hesitations	6.9 / 0.7	20.3 / 1.1	16.7 / 1.1
Agent package offers	0.84	4.28	3.73
Budget stated (%) / Task success (%)	63.7% / 92.1%	70.7% / 94.2%	75.4% / 95.3%

Table D4

Primary Archetype Assignment and Centroid Similarity Across Partitions and MultiWOZ 2.2

Sample	*N*	Centroid cosine *M* (*SD*)	Direct Transactional (%)	Constraint Negotiator (%)	Guided Explorer (%)
Frames golden	933	.895 (.042)	25.7	24.1	50.2
Frames held-out	401	.894 (.042)	29.4	20.2	50.4
MultiWOZ 2.2	1,499	.805 (.037)	0.9	15.1	84.0

Table D5

Stability, Routing Agreement, and Outcome Association of Hybrid Archetypes by Conversation Length

User messages used (*M*)	Dialogues shortened (%)	Bootstrap ARI	Core members (%)	Smallest cluster (%)	Raw L1 κ	Blocks router κ	Outcome *V*
First 3 messages	94.7	.89	92.7	18.3	.72	.359	.16
First 5 messages	78.2	.84	77.1	20.3	.38	.535	.27
First 7 messages	53.2	.85	89.9	29.7	.25	.722	.38
First 9 messages	30.7	.89	93.1	25.5	.24	.958	.45
Whole conversation	0.0	.62	53.7	7.9	.22	—	.57

Note. Raw L1 κ uses unprojected 5-turn features (step2g); Blocks router κ projects prefix features into the exact golden Blocks space (RefinedArchetypeRouter).

Table D6

Ablation of the Refined Archetype Definition at M = 9 User Messages (k = 3)

Representation configuration	ARI (whole)	ARI (9 msgs)	Core members (%)	Outcome *V*	ARI vs. adopted
Gemini embedding + Gemini behavior profile (adopted)	.56	.87	89.2	.45	1.00
MiniLM embedding + Gemini behavior profile	.69	.85	87.7	.40	.33
Gemini behavior profile only	.73	.85	97.6	.35	.17
Gemini embedding only	.91	.91	92.0	.34	.45
MiniLM embedding only	.68	.72	72.1	.27	.14
MiniLM embedding + dialogue-act features (no LLM)	.61	.76	88.0	.45	.16

Figure D1

Cluster-Number Diagnostics on Golden Partition

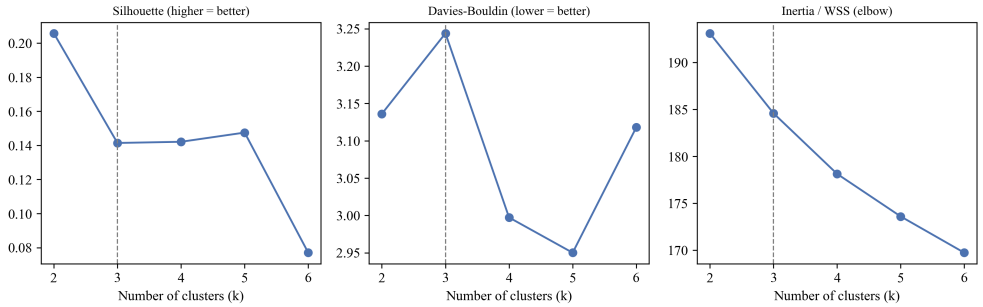


Figure D2

Silhouette Coefficients Across 159 Subsample Runs by Algorithm

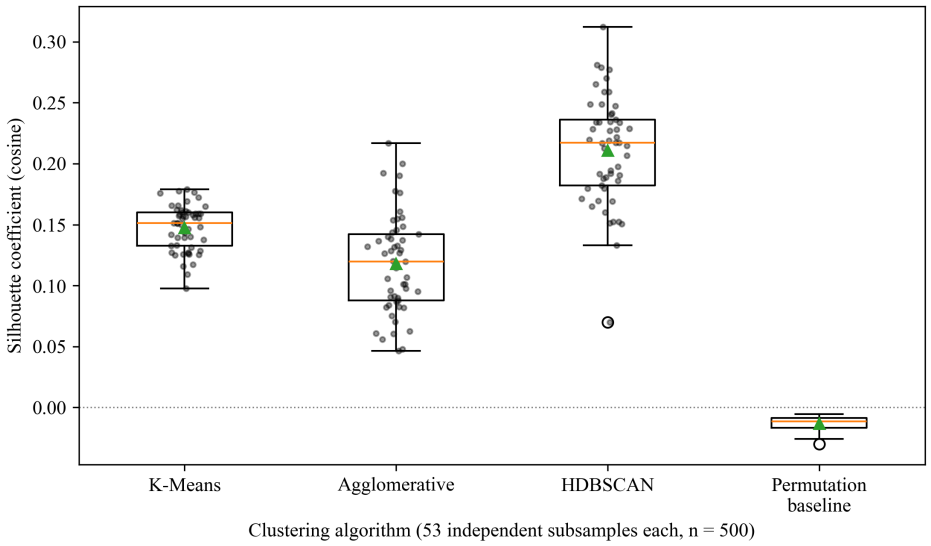


Figure D3

UMAP Projection of Frames Archetypes and MultiWOZ 2.2 External Transfer Check

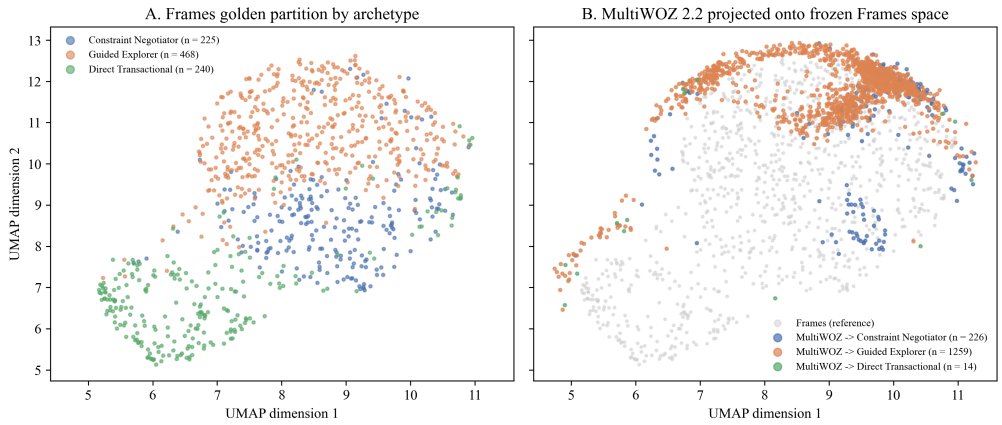
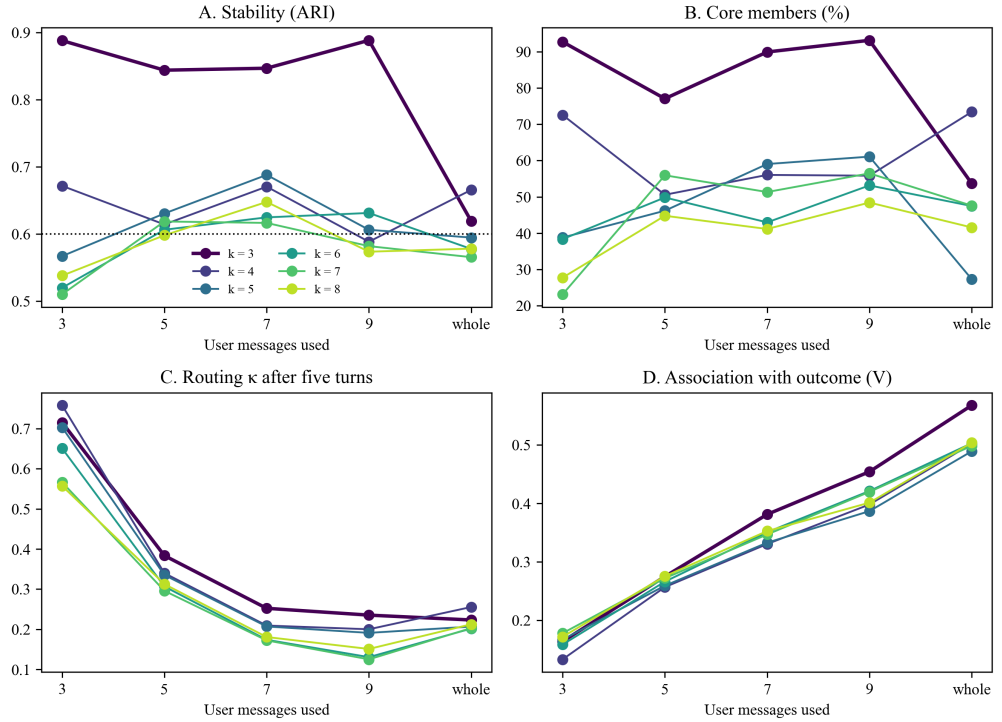


Figure D4

Hybrid Archetype Stability, Routing, and Outcome Association Across Truncation Horizons



Appendix E

Supplementary EDA, RQ2 Pairwise Contrasts, and RQ3/RQ4 Groundwork Diagnostics

Tables E1–E5 and Figures E1–E3 provide the supplementary slot-completeness table, multivariate PCA loadings, Holm-adjusted pairwise contrasts for RQ2, corpus rule coverage audit for RQ3, and human turn-economy baseline for RQ4.

Table E1

Observed Canonical Slot Completeness by Corpus

Canonical slot	Schema tier	Microsoft Frames (%)	MultiWOZ 2.2 (%)
origin city	Tier 1	99.4	70.8
destination city	Tier 1	99.0	71.4
departure date	Tier 1	84.9	68.6
return date	Tier 1	70.8	0.0
duration nights	Tier 1	61.8	27.0
traveler count adults	Tier 1	63.9	54.0
traveler count children	Tier 1	18.4	0.0
budget amount	Tier 1	72.9	24.1
accommodation tier	Tier 1	72.7	19.7
visa processing window days	Tier 2	0.0	0.0
passport expiry months	Tier 2	0.0	0.0
discount applied percentage	Tier 2	0.0	0.0

Table E2

Multivariate Principal Component Loadings of Structural and Behavioral Features (Frames)

Variable	PC1 loading (40.8% var.)	PC2 loading (17.2% var.)	PC3 loading (10.6% var.)
Total Turns	.393	.166	−.234
Slot Revisions	.424	.195	−.064
Agent Offers	.361	.292	−.078
User Words/Turn	.025	−.087	.888
Dest. Flexibility	.327	−.182	.135
Exploration Breadth	.417	−.108	.101
Decisiveness	−.259	.503	.028
Revision Tendency	.297	−.460	.025
Question Asking	.301	.248	.027
Politeness	.089	.520	.341

Table E3

Holm-Adjusted Paired Contrasts for Refined RQ2 Outcomes (N = 33)

Outcome metric	Pairwise contrast	Mean diff.	*t*	*df*	*p* (Holm)	d_z
Refined: SBERT style cosine	Cluster-Adaptive vs. Static	.144	4.24	32	< .001	0.74
Refined: SBERT style cosine	Cluster-Adaptive vs. Zero-Shot	.067	2.36	32	.049	0.41
Refined: SBERT style cosine	Static vs. Zero-Shot	−.077	−2.20	32	.049	−0.38
Refined: Gemini style cosine	Cluster-Adaptive vs. Zero-Shot	.027	1.20	32	.713	0.21
Refined: Gemini style cosine	Static vs. Zero-Shot	.024	0.79	32	.865	0.14
Refined: Gemini style cosine	Cluster-Adaptive vs. Static	.004	0.13	32	.895	0.02
Refined: BERTScore F1	Cluster-Adaptive vs. Static	.056	1.45	32	.472	0.25
Refined: BERTScore F1	Cluster-Adaptive vs. Zero-Shot	.025	0.85	32	.804	0.15
Refined: BERTScore F1	Static vs. Zero-Shot	−.031	−0.75	32	.804	−0.13
Refined: Pacing error (words)	Cluster-Adaptive vs. Zero-Shot	−6.25	−3.49	32	.004	−0.61
Refined: Pacing error (words)	Cluster-Adaptive vs. Static	−2.46	−2.66	32	.024	−0.46
Refined: Pacing error (words)	Static vs. Zero-Shot	−3.79	−1.82	32	.078	−0.32
Refined: Response length (words)	Cluster-Adaptive vs. Static	6.82	7.00	32	< .001	1.22
Refined: Response length (words)	Static vs. Zero-Shot	−10.91	−4.29	32	< .001	−0.75
Refined: Response length (words)	Cluster-Adaptive vs. Zero-Shot	−4.09	−1.84	32	.075	−0.32

Table E4

Empirical Corpus Audit of Policy Rule Coverage in Microsoft Frames (N = 1,334)

Declarative rule	Required canonical slot	Slot observed (%)	Dialogues triggering (*n*)	Triggering (%)
pax capacity	traveler count adults	24.1	54	4.05
budget minimum	budget amount	65.1	266	19.94
visa window	visa processing window days	0.0	0	0.00
passport validity	passport expiry months	0.0	0	0.00
discount threshold	discount applied percentage	0.0	0	0.00

Table E5

Human Wizard Baseline of Turn Economy and Redundant Elicitation by Refined Archetype (Frames)

Refined user archetype	*n*	Turns *M* (*SD*)	RER *M* (*SD*)	Wizard request turns *M*
Decisive Booker	595	14.56 (4.52)	.014 (.047)	1.44
Open Explorer	337	20.75 (7.07)	.016 (.049)	1.82
Rigid-Constraint Seeker	402	10.60 (5.59)	.017 (.061)	1.27

Figure E1

Observed Slot Completeness by Corpus

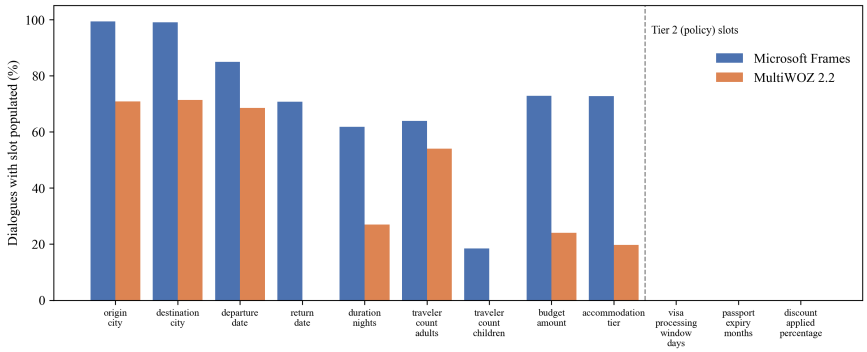


Figure E2

Primary RQ2 Results on Unrefined Whole-Dialogue Archetypes

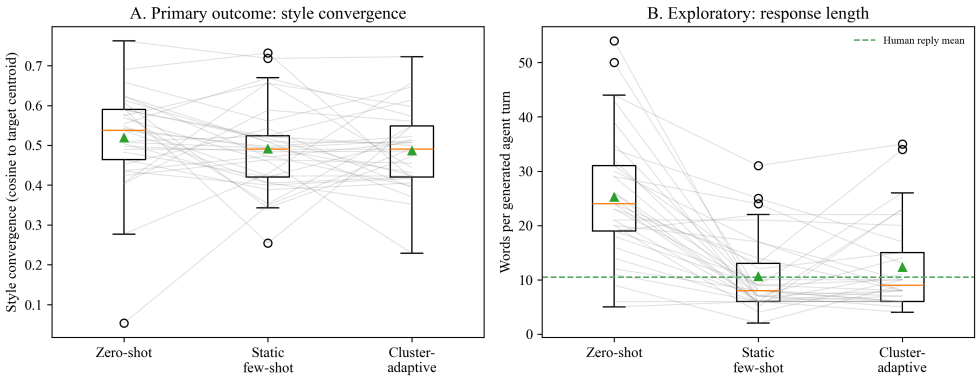


Figure E3

Human Wizard Baseline of Turn Economy and Redundant Elicitation by Refined Archetype

